CISS350: Data Structures and Advanced Algorithms
Quiz q01

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For each of the functions below, you are only allowed to use one loop.

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Q1. Write a function `replace()` that replaces a character in a C-string with another. For instance suppose you have

```
char x[1024] = "helloworld";
```

If you call

```
replace(x, 'o', 'a');
```

and then you print `x`, the output is

```
hellawarld
```

Answer:

```
#include <iostream>

void replace(char * x, char src, char dest)
{
  for(int i = 0; x[i] != '\0'; i++)
  {
    if (x[i] == src)
        x[i] = dest;
  }
}

int main()
{
    char x[1024];
    char src, dest;
    std::cin >> x >> src >> dest;
    replace(x, src, dest);
    std::cout << '[' << x << "]\n";
    return 0;
}
```

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Q2. Write a function `remove()` that removes all occurrences of a character from a C-string. For instance suppose you have

```
char x[1024] = "helloworld";
```

If you call

```
remove(x, 'o');
```

and then you print `x`, the output is

```
hellwrld
```

Answer:

```
#include <iostream>

void remove(char * x, char c)
{
  int j = 0;
  for(int i = 0; x[i] != '\0'; i++)
  {
    if (x[i] != c)
    {
      x[j++] = x[i];
    }
  }
  x[j] = '\0';
}

int main()
{
    char x[1024];
    char c;
    std::cin >> x >> c;
    remove(x, c);
    std::cout << '[' << x << "]\n";
    return 0;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.